

Vaibhav Saini

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Summary

Early career AI and Software Engineer with hands on experience building production RAG systems, LLM cost routers, and async API backends from scratch. I care about latency, failure modes, and whether AI output can actually be trusted. I move fast, take full ownership, and have independently designed and deployed AI systems end to end. On OPT, looking for a full-time role.

Experience: Software Engineer | Badoliasoft (Oct 2023 - Jun 2024)

- Implemented JWT auth with HMACSHA512 hashing, 3 tier RBAC, and token blacklisting, securing a fintech platform for 100+ users with zero unauthorized access incidents post-deployment.
- Built real time notification pipeline using SignalR with RabbitMQ fanout exchange fallback; integrated Azure Blob storage with CDN, reducing average file delivery latency by 40%.
- Configured IP-based rate limiting, global exception middleware with structured logging, and Redis distributed caching, reducing database load by 35% and improving API reliability under peak traffic.
- Contributed to a full-stack Angular + .NET 8 platform (over 200 C# files, 10 REST controllers) spanning user management, messaging, and project workflows.
- Operated as a core contributor on a small team building a greenfield fintech platform from scratch, owning auth, notifications, and caching systems across the full delivery cycle.

Technical Skills

- **Language & Frameworks:** Python, Java, Go, FastAPI, TypeScript, Node.js, React, .NET, Angular
- **AI & Machine Learning:** LLMs, RAG, Embeddings, Sentence-Transformers, Fine-Tuning, Prompt Engineering, Vector Search, LangChain
- **Cloud & DevOps:** AWS, Azure, Docker, CI/CD
- **Databases:** PostgreSQL, DynamoDB, Pinecone, ChromaDB, SQL Server, Redis, Vector Database
- **Security:** Splunk, Wireshark, JWT Auth, RBAC, OWASP, Monitoring

Projects

SecRAG Pro: Production RAG system with Intelligent LLM Cost Routing | FastAPI, ChromaDB, scikit-learn, Sentence Transformers, OpenAI, Anthropic, Ollama, Docker

- Merged two independently built systems; a hybrid RAG pipeline (SecRAG) and an LLM cost router (LLM Cost Autopilot) into a unified production backend, eliminating redundant inference calls and centralizing routing logic across GPT-4o, GPT-4o Mini, Claude Sonnet, Claude Haiku, and local Llama 3.2.
- Achieved 62.3% cost reduction vs. all GPT-4o routing across 513 test queries with 0 failures; routed 41% of queries to free local inference via Ollama without measurable quality degradation.
- Built hybrid BM25 plus dense vector retrieval fused via Reciprocal Rank Fusion with a cross encoder reranker (ms-marco-MiniLM-L-6-v2) cutting top 20 candidates to top 5, achieving 4.15/5.0 average answer quality on a 48-question golden eval set.
- Validated answer trustworthiness via LLM as judge verification that checks every claim against source chunks, achieving 99.6% citation accuracy which is critical for high stakes document queries.
- Returned per request routing metadata alongside every answer with model used, cost, cost saved, and complexity tier; enabling full cost observability across all providers.
- Currently researching and extending the system into LegalAdaptRAG; an intent aware adaptive retrieval pipeline for legal document understanding with added intent classification (5 query types), per intent retrieval strategy selection, citation verification, and confidence scoring targeted at underserved users who cannot afford legal representation.

Education

Master of Science in Information Technology (Security): May 2026

Arizona State University, Tempe, AZ (GPA: 3.97 / 4.0)

Bachelor of Engineering in Computer Science (Data Science): June 2023

MMDU, Ambala, Haryana, India GPA: 7.62 / 10.0

Certifications

- **Generative AI and LLMs:** DeepLearning.AI (February 2026)
- **Machine Learning Specialization:** DeepLearning.AI (January 2026)
- **AWS Certified Cloud Practitioner:** Amazon Web Services (July 2025)
- **AWS Academy Cloud Architect and Solutions Engineer:** Amazon Web Services (April 2025)